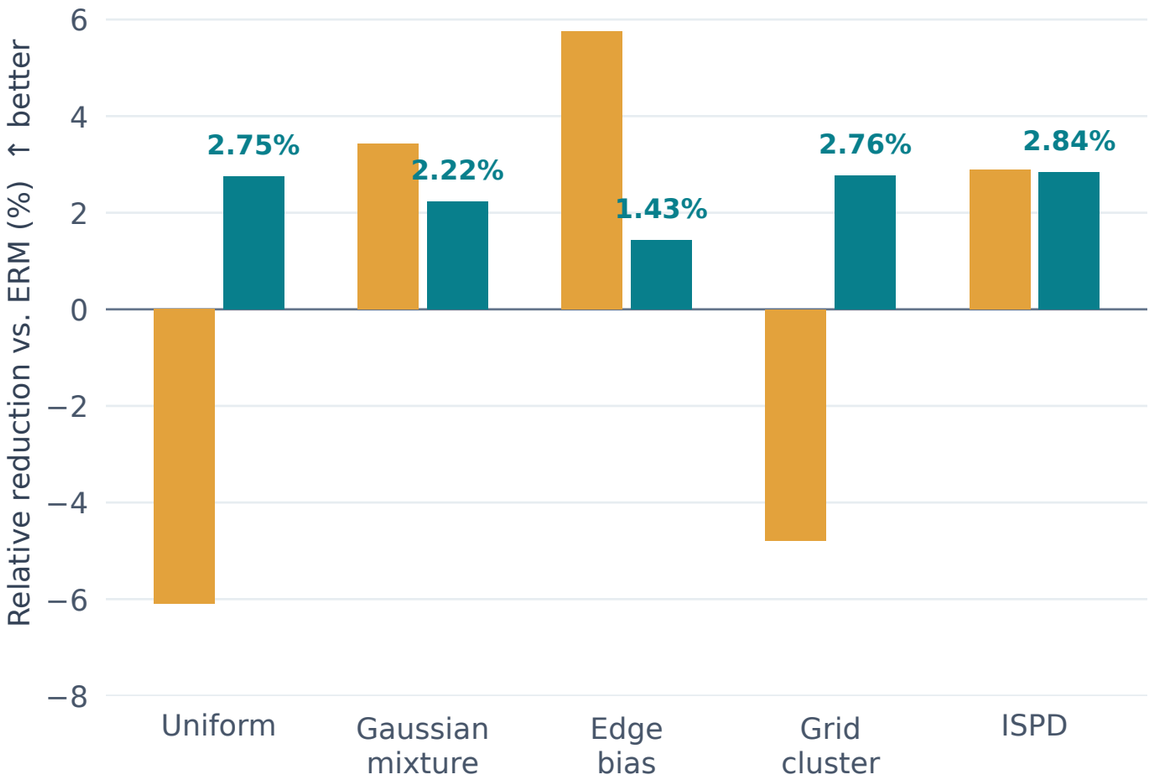
RESULTS AT A GLANCE

WDRO reduces mean regret by 1.43–2.84% versus ERM

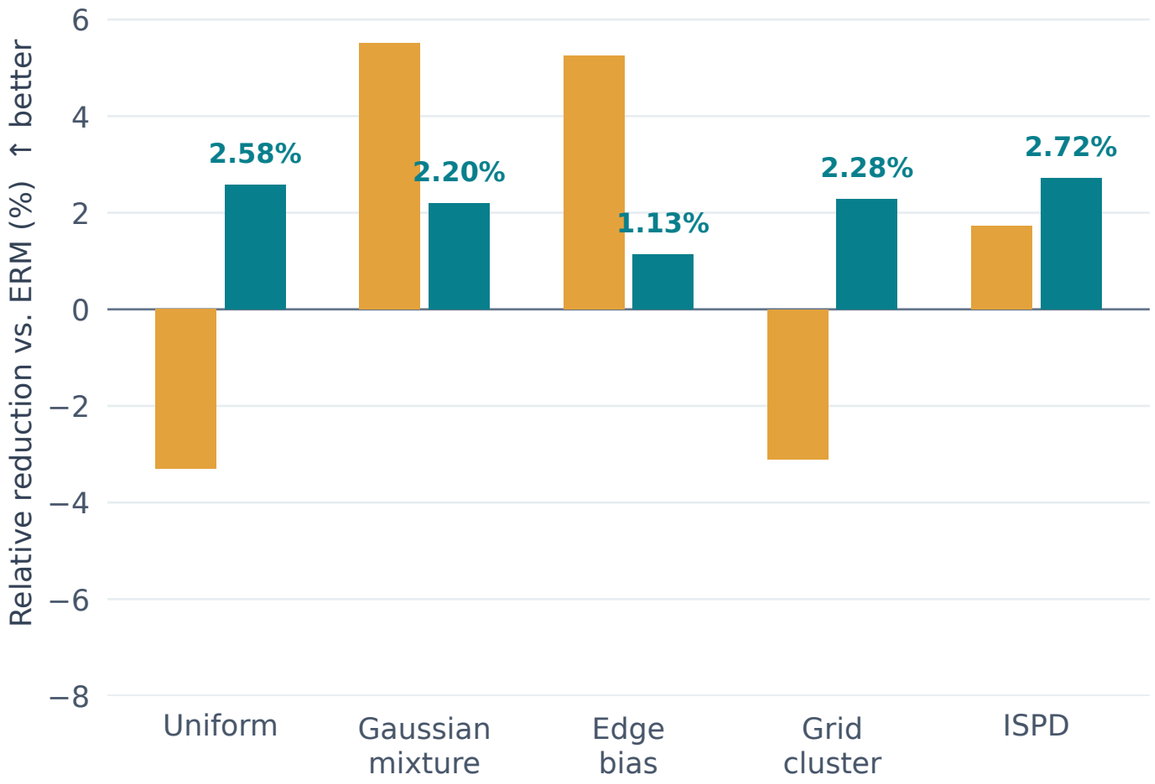
All four synthetic distributions and the ISPD test set | 3 training seeds | frozen model evaluation

Mixed ERM WDRO-REST

Mean regret reduction



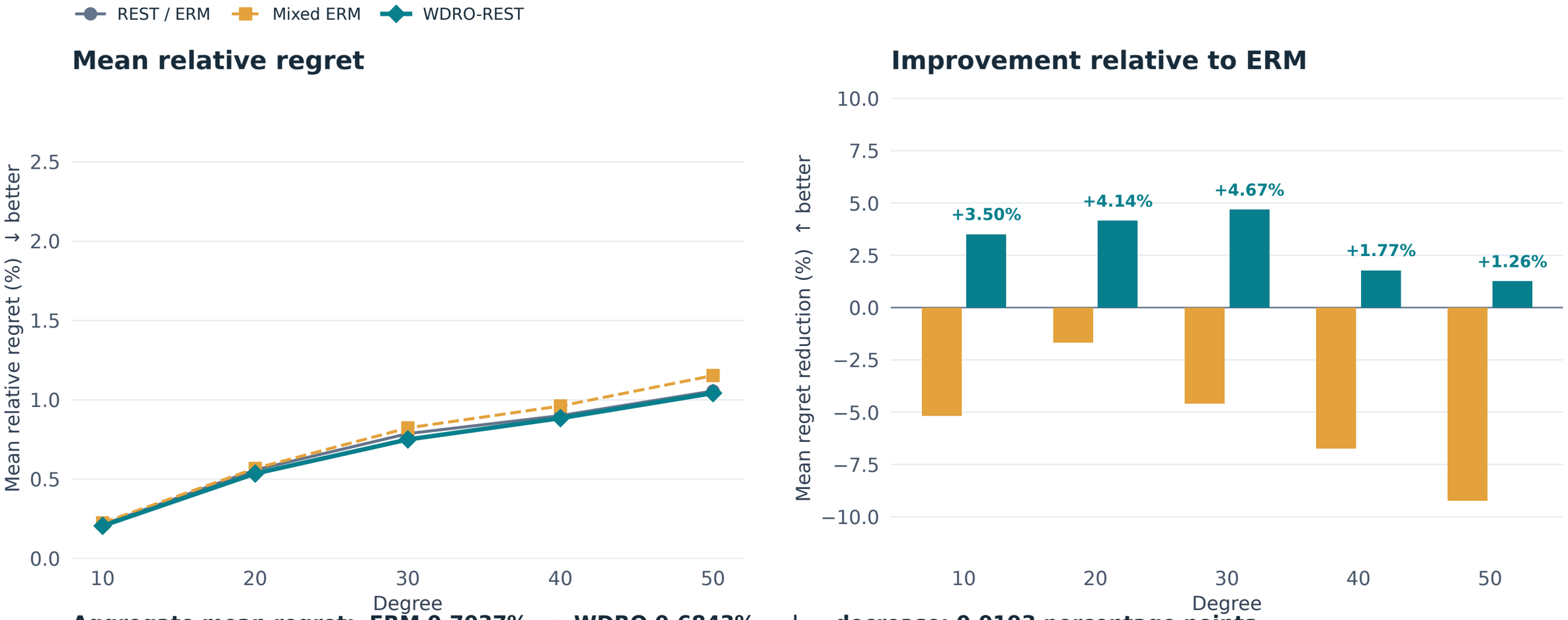
P95 regret reduction



Positive = lower regret than ERM; negative = higher regret. Reduction = (ERM − method) / ERM × 100.
Synthetic: equal degree/seed weights; P95 computed within each degree/seed, then averaged. ISPD: pooled nets per seed, then seed average.
Mixed ERM is stronger on Gaussian mixture and edge bias; these results do not show WDRO superiority over every baseline.

Uniform: 2.75% lower mean regret with WDRO

REST / ERM, Mixed ERM and WDRO-REST | 3-seed averages | degree = number of pins

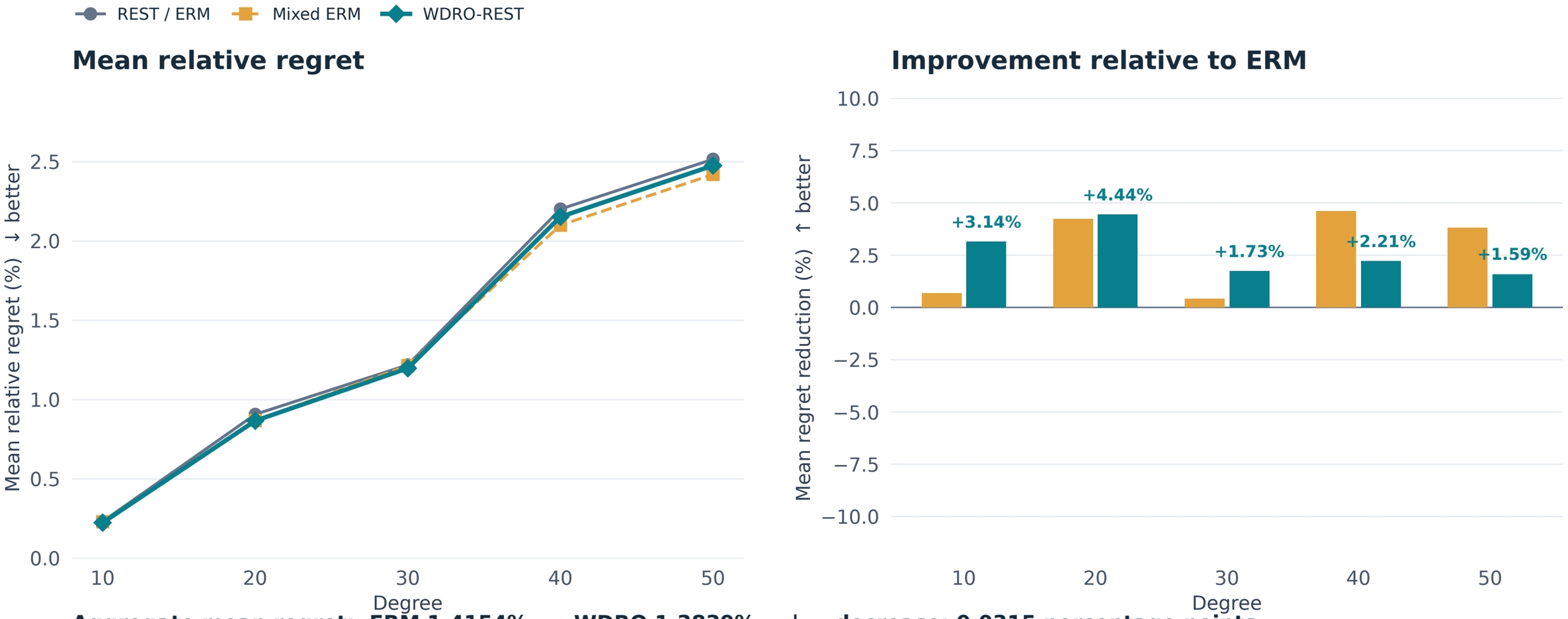


Aggregate mean regret: ERM 0.7037% → WDRO 0.6843% | decrease: 0.0193 percentage points

10,000 shared test nets per degree; all 5 degrees and all 3 seeds included. Aggregate uses equal degree/seed weights.
Regret = (REST length − reference length) / reference length. Bars: (ERM − method) / ERM × 100; positive = better.
Fixed axis ranges across dataset figures; point estimates only. Source: precision_v2 stage 6 / stage 7 frozen result tables.

Gaussian mixture: 2.22% lower mean regret with WDRO

REST / ERM, Mixed ERM and WDRO-REST | 3-seed averages | degree = number of pins

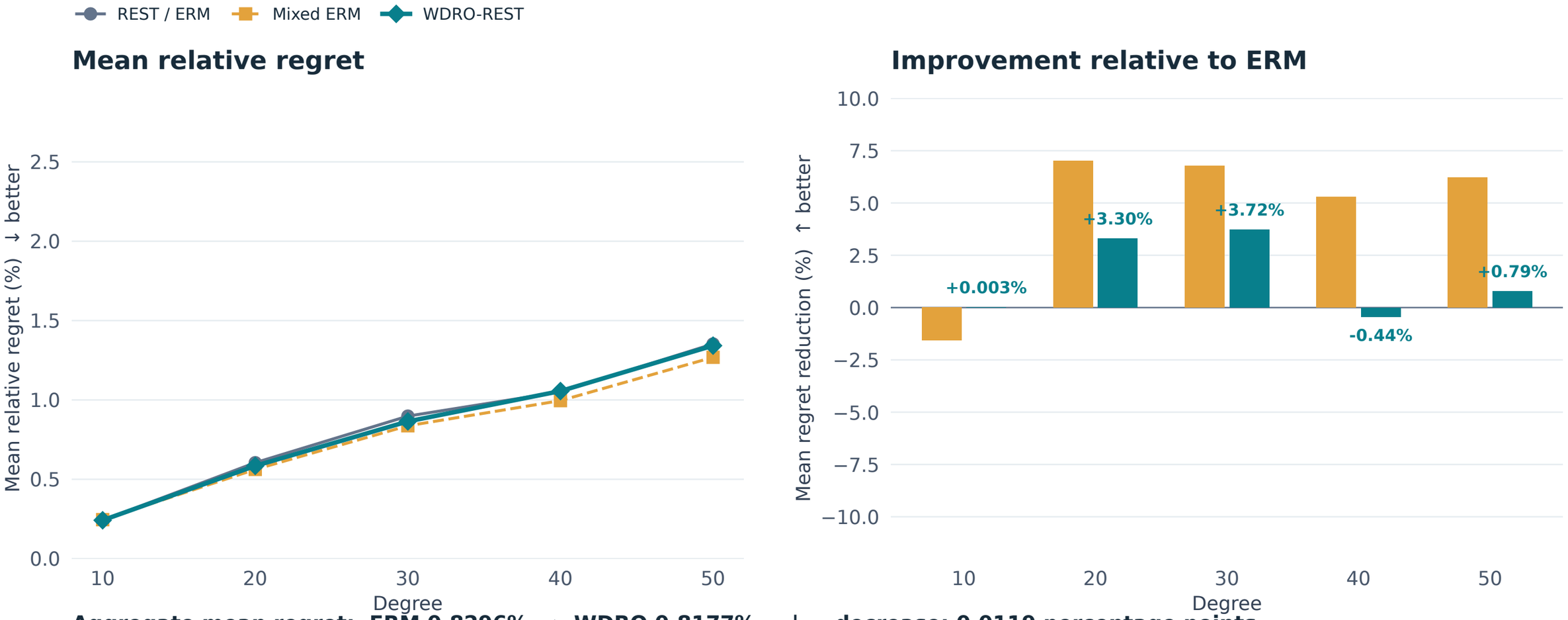


Aggregate mean regret: ERM 1.4154% → WDRO 1.3839% | decrease: 0.0315 percentage points

10,000 shared test nets per degree; all 5 degrees and all 3 seeds included. Aggregate uses equal degree/seed weights.
Regret = (REST length − reference length) / reference length. Bars: (ERM − method) / ERM × 100; positive = better.
Fixed axis ranges across dataset figures; point estimates only. Source: precision_v2 stage 6 / stage 7 frozen result tables.

Edge bias: 1.43% lower mean regret with WDRO

REST / ERM, Mixed ERM and WDRO-REST | 3-seed averages | degree = number of pins

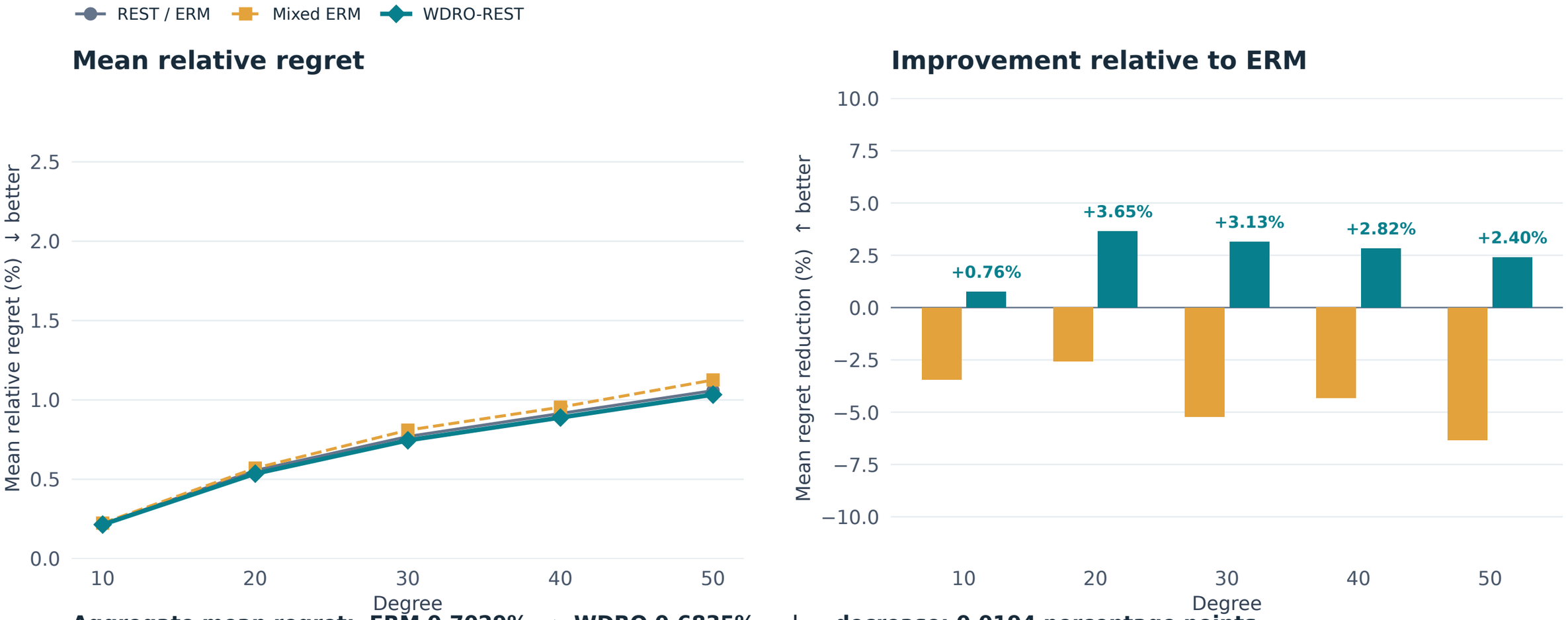


Aggregate mean regret: ERM 0.8296% → WDRO 0.8177% | decrease: 0.0119 percentage points

10,000 shared test nets per degree; all 5 degrees and all 3 seeds included. Aggregate uses equal degree/seed weights.
Regret = (REST length − reference length) / reference length. Bars: (ERM − method) / ERM × 100; positive = better.
Fixed axis ranges across dataset figures; point estimates only. Source: precision_v2 stage 6 / stage 7 frozen result tables.

Grid cluster: 2.76% lower mean regret with WDRO

REST / ERM, Mixed ERM and WDRO-REST | 3-seed averages | degree = number of pins

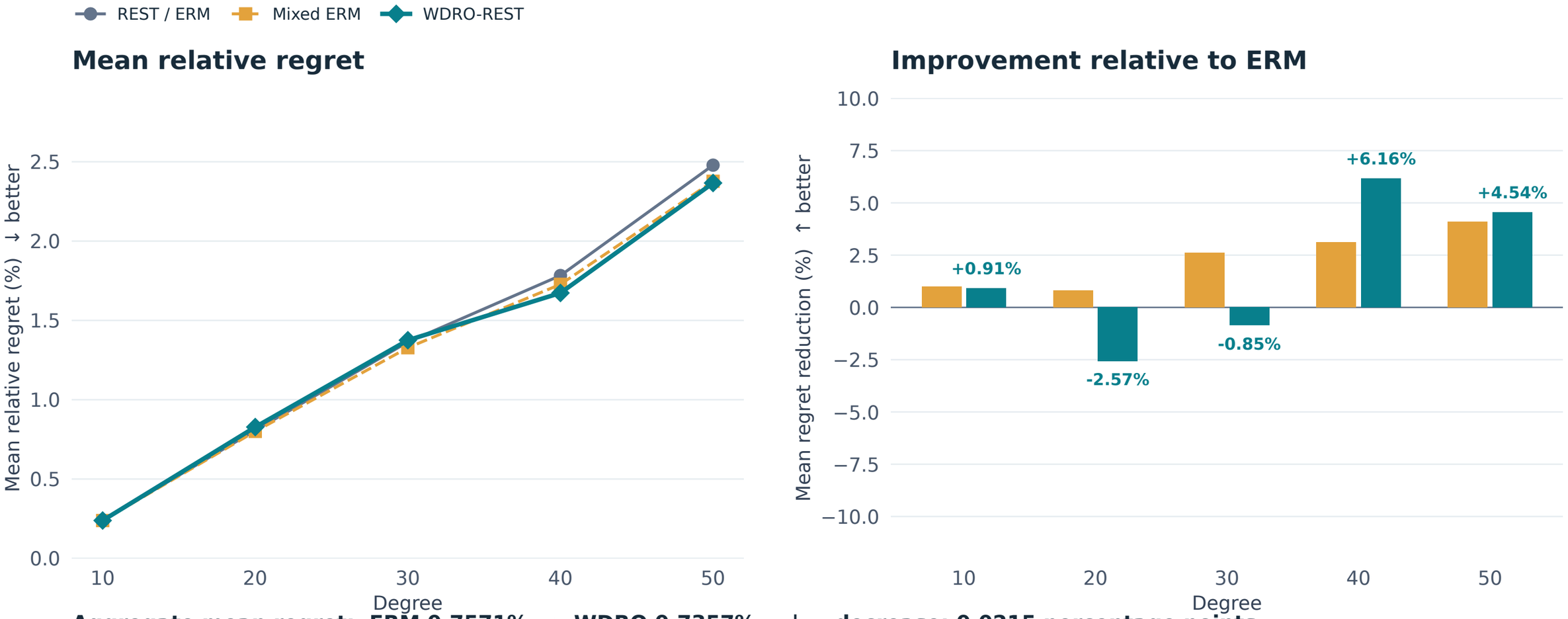


Aggregate mean regret: ERM 0.7029% → WDRO 0.6835% | decrease: 0.0194 percentage points

10,000 shared test nets per degree; all 5 degrees and all 3 seeds included. Aggregate uses equal degree/seed weights.
Regret = (REST length − reference length) / reference length. Bars: (ERM − method) / ERM × 100; positive = better.
Fixed axis ranges across dataset figures; point estimates only. Source: precision_v2 stage 6 / stage 7 frozen result tables.

ISPD: 2.84% lower mean regret with WDRO

REST / ERM, Mixed ERM and WDRO-REST | 3-seed averages | degree = number of pins



Aggregate mean regret: ERM 0.7571% → WDRO 0.7357% | decrease: 0.0215 percentage points

ISPD: 22,498 nets from 20 designs; no ISPD training. Aggregate uses net weights; degree panels average 3 seeds.
Regret = (REST length − reference length) / reference length. Bars: (ERM − method) / ERM × 100; positive = better.
Fixed axis ranges across dataset figures; point estimates only. Source: precision_v2 stage 6 / stage 7 frozen result tables.